Practice workbook · Explain genetic and environmental causes of variation

Whether a feature varies because of inherited alleles, surroundings or both.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EXPLAIN-GENETIC-AND-ENVIRONMENTAL-VARIATION

1. Question 3 · part d · 3 marks

November 2021 Paper 2

Worked-example retry · Attempt part d only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Sweating is normal when we are hot. It is part of our natural cooling mechanism. If you sweat visibly when you do not need to cool down, it can be a problem.

The sweating process is controlled by the brain, which sends signals down the nerves to the sweat glands.

Sweating is crucial to survival. We sweat to control body temperature and the sweat glands in our armpits can produce several litres of sweat in 24 hours. How much we sweat varies hugely from individual to individual and even from day to day.

Hyperhidrosis occurs when some areas of the body start sweating excessively. This affects about 1% of the population. There are two main types. The most common type is primary focal hyperhidrosis, which affects many parts of the body. There is no known cause, but it usually begins in childhood and often runs in families. Another type is secondary hyperhidrosis, which often doesn't begin until after the teenage years, and usually has an underlying medical cause. These causes can include diabetes, menopause or chronic infection. It can also be a result of eating certain foods. Stress can play a major role too.

(a) Which part of the brain controls sweating (line 4)?

(1)

- ☒ **A** cerebellum
- ☒ **B** cerebral hemispheres
- ☒ **C** hypothalamus
- ☒ **D** pituitary gland

(b) Which type of nerve sends signals to the sweat glands (lines 4 and 5)?

(1)

- ☒ **A** connector
- ☒ **B** motor
- ☒ **C** relay
- ☒ **D** sensory

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1. Question 3 · part d · 3 marks

November 2021 Paper 2

Worked-example retry · Attempt part d only.

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(c) Describe how sweating controls body temperature (lines 6 and 7).

(3)

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(d) Explain the evidence in the passage that indicates primary focal hyperhidrosis is a genetic condition.

(3)

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(e) Secondary hyperhidrosis can be caused by diabetes.

Give one risk factor for diabetes (line 16).

(1)

1. Question 3 · part d · 3 marks · Working and answer

Working and answer

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2. Question 3 · part b · 3 marks

November 2020 Paper 1

Further practice · Attempt part b only.

3 There are many factors that affect the height of a human.

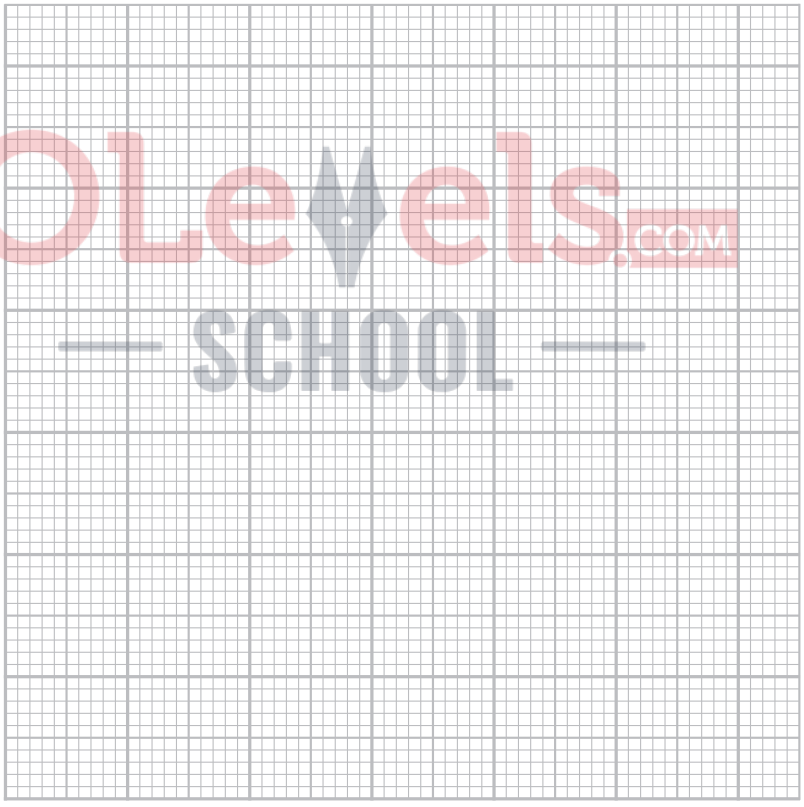
(a) The heights of a group of students are measured.

The table shows the results.

Height in cm	150–151	152–153	154–155	156–157	158–159	> 160
Number of students	4	14	25	20	10	4

(i) Plot a bar chart of the results.

(5)



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2. Question 3 · part b · 3 marks

November 2020 Paper 1

Further practice · Attempt part b only.

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(ii) Describe the trend shown in this bar chart.

(2)

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(b) State three factors that could affect the heights of the students.

(3)

1 .....

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2 .....

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3 .....

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(Total for Question 3 = 10 marks)

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2. Question 3 · part b · 3 marks · Working and answer

Working and answer

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3. Question 5 · part d · 2 marks

June 2022 Paper 2R

Further practice · Attempt part d only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Milk contains a carbohydrate called lactose. Lactose is not absorbed by the epithelium of the ileum or duodenum.

Lactose is a disaccharide which is broken down into glucose and galactose by the enzyme lactase. Babies produce large amounts of lactase but in some people the gene for producing lactase is switched off as they get older. When this gene is switched off people then become lactose-intolerant which means they cannot digest any lactose that is taken into the gut.

The number of people with lactose intolerance differs across the world. People in some countries are often tolerant because their ancestors have evolved tolerance because of their diet.

In a lactose intolerant person, any lactose taken into the gut will collect and ferment, forming methane and carbon dioxide plus other soluble products. These other soluble products lower the water potential in the gut.

Symptoms of lactose intolerance include diarrhoea, calcium deficiency and vitamin D deficiency.

(a) Suggest what is meant by the term disaccharide. (lines 3 to 4) **(2)**

(b) Give a reason why lactose is not absorbed by the epithelium. (lines 1 to 2) **(1)**

(c) (i) Explain what is meant by a gene being switched off. (lines 5 to 6) **(3)**

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3. Question 5 · part d · 2 marks

June 2022 Paper 2R

Further practice · Attempt part d only.

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(ii) Describe the effects on a person when the gene for producing lactase is switched off. (lines 3 to 7)

(2)

(d) Suggest how people in some parts of the world have evolved tolerance because of their diet. (lines 8 to 10)

(2)

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(e) Explain how the symptoms of lactose intolerance are caused. (lines 11 to 15)

(4)

(Total for Question 5 = 14 marks)

3. Question 5 · part d · 2 marks · Working and answer

Working and answer

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